

1. Introduction & Safety Overview

Purpose of the System यह unit **Hydrogen gas** को ultra-dry और oxygen-free बनाता है।

- **Oxygen removal** → Explosion risk रोकने के लिए ($H_2 + O_2$ explosive mixture बनाता है)
- **Moisture removal** → Dew Point -60°C या उससे नीचे तक लाना, corrosion रोकना, downstream equipment (compressors, fuel cells, processes) की protection करना।

Final Product Spec (Typical):

- Oxygen: $< 10\text{--}20$ ppm
- Dew Point: -60°C to -70°C or better
- Purity: 99.999%+ (application dependent)

Safety First (Plant Engineer Mindset):

- Hydrogen highly flammable & explosive → LEL 4%, UEL 75% in air
- Deoxo reaction exothermic → Runaway temperature possible
- High pressure & temperature in towers
- Always follow LOTO, purging with N_2 before maintenance
- Catalyst handling: Pd/Pt based → sensitive to poisons (S, CO, organics)

Overall Process Flow (Big Picture): Raw H_2 (with $O_2 + H_2O$) → **Deoxo Unit** → **Water Cooling** → **Moisture Separator** → **3-Tower Dryer System** → Dry Pure H_2

2. Deoxo Unit – Oxygen Removal (Critical Safety Section)

Working Principle: Catalytic recombination: $2H_2 + O_2 \rightarrow 2H_2O$ ($\Delta H = \text{exothermic}$)

Key Components:

- Catalyst: Palladium (Pd) or Platinum (Pt) on Alumina support (most common)
- Operating Temperature: $150\text{--}300^\circ\text{C}$ (inlet often preheated)
- Reaction starts at lower temp but optimal at higher temp for complete conversion

Process Steps:

1. Hydrogen gas enters Deoxo vessel
2. Catalyst bed में O_2 react करके steam बनाता है

3. Outlet gas: Hot + wet (steam) + very low O₂

Important Parameters to Monitor (Log Sheet):

- Deoxo Top & Bottom Temperature (ΔT indicates reaction zone)
- Oxygen Analyzer (downstream) – alarm if >20–50 ppm
- Inlet/Outlet Pressure

Practical Logic:

- Low temperature → Incomplete reaction → O₂ breakthrough → Dryer desiccant damage + safety risk
- High temperature → Catalyst sintering or damage
- Catalyst Poisoning: Sulfur compounds, CO, hydrocarbons → activity loss

Startup/Shutdown Best Practices:

- Gradual heating (50–80°F/hr) to avoid thermal shock
- Purge with inert (N₂) before introducing H₂
- Monitor temperature rise due to exothermic reaction

Common Issues & Actions:

- O₂ high → Check catalyst activity, increase temp, or replace catalyst
 - Temperature excursion → Reduce inlet O₂ or flow rate
-

3. Water Cooling & Moisture Separator

After Deoxo: Gas बहुत hot (150–300°C) + steam के साथ निकलता है।

Water Cooling Section:

- Shell & Tube or Air Cooler type heat exchanger
- Cooling water / chilled water से gas को 30–50°C तक ठंडा करते हैं
- Steam condense होकर water droplets बनते हैं

Moisture Separator:

- Knock-out drum or demister pad वाला vessel
- Auto drain valve (float type) या manual purge
- Water को bottom से drain करते हैं

Why Critical?

- Hot wet gas direct dryer में गया → Desiccant (Molecular Sieve) immediately saturate or damage हो जाएगा
- Carryover water → Tower life drastically reduce

Monitoring & Problems:

- Cooling water flow & inlet/outlet temp difference
- Separator level (high level alarm → carryover risk)
- Issues: Scale in cooler → poor heat transfer; Drain choke → water backup

Engineer Tip: Regularly check cooling water quality to prevent scaling. Inspect demister pads during shutdown.

4. 3-Tower Dryer System (TSA – Temperature Swing Adsorption)

Why 3 Towers? Continuous operation के लिए। 2-tower system में regeneration के दौरान flow interrupt हो सकता है, 3-tower staggered cycle देता है smooth operation।

Tower Roles (Cyclic – PLC Controlled):

Tower	Current Mode	What Happens Inside
1	Drying (Adsorption)	Wet H ₂ passes through → Moisture adsorbed by desiccant → Dry H ₂ out
2	Heating (Regeneration)	Hot dry purge gas (H ₂ or N ₂) passes → Desorbs moisture → Wet purge out to vent/drain
3	Cooling	Cold dry gas passes → Tower cooled down to adsorption temp

Typical Cycle Time: 4–8 hours per step (total cycle 12–24 hours), depending on load & design. PLC automatically switches valves based on time or dew point breakthrough.

Desiccant Used:

- **Molecular Sieve (Zeolite 3A or 4A)** – Most common for deep drying (low dew point)
- Sometimes Activated Alumina (cheaper but higher dew point)

Adsorption Mechanism: Desiccant pores में water molecules trapped (physical adsorption). Capacity depends on temp & partial pressure of water.

Detailed Modes:

1. Drying Mode (Adsorption):

- Inlet wet gas (after separator) top से enters
- Moisture adsorbed → Outlet dew point बहुत low ($-60^{\circ}\text{C}+$)
- Monitor: Outlet Dew Point Analyzer (breakthrough when rises)

2. Heating/Regeneration Mode:

- Dry purge gas (usually slipstream of product H_2 or N_2) को heater से $200\text{--}315^{\circ}\text{C}$ तक गर्म करें
- Hot gas counter-current या co-current flow से tower pass
- Adsorbed water desorbs (vaporizes) और purge gas के साथ बाहर निकलता है
- Regeneration temp & flow rate critical for complete moisture removal

3. Cooling Mode:

- Regeneration के बाद tower hot रहता है ($200^{\circ}\text{C}+$)
- Cold dry gas (product H_2 slip) से ठंडा करें (ambient या cooled)
- Why? Hot bed moisture को अच्छे से adsorb नहीं करता → immediate breakthrough

Valve System:

- Multiple automatic valves: Inlet, Outlet, Heating inlet/outlet, Cooling, Purge, Depressurization
- PLC sequence: Safety interlocks (e.g., no heating without purge flow)
- Leakage in valves → cross contamination या poor performance

Engineer Note: Regeneration gas consumption typically 10–20% of main flow. Optimize to save H_2 .

5. Instrumentation & Control (Log Sheet Reading)

Key Instruments (Hourly Log Sheet Values):

- **Pressures:** H_2 inlet, N_2 (for purge/regeneration), Air (instrument)
- **Temperatures:** Deoxo Top/Bottom, Each Tower (Top, Middle, Bottom – profile important)
- **Dew Point Analyzer:** Final outlet (most critical quality parameter)

- **Oxygen Analyzer:** Post-Deoxo
- **Flow:** H₂ consumption, Purge flow
- **Others:** Differential pressure across towers (bed fouling indicator)

How to Read Log Sheet Practically:

- Tower temps देखकर समझो कौन सा mode में है (e.g., Tower 1 low & stable temp = drying; Tower 2 rising to 250°C = heating)
- Dew point sudden rise → Tower saturated or valve issue
- Deoxo ΔT abnormal → Reaction problem

Control: Mostly PLC based with automatic cycle switching. Manual override for maintenance.

6. Operating Procedures (Step-by-Step)

Normal Operation:

1. Confirm all utilities (cooling water, instrument air, power)
2. Start Deoxo (gradual heat + H₂ flow)
3. Cooling & Separator online
4. Dryer towers in cycle (one drying, one heating, one cooling)
5. Monitor dew point < -60°C & O₂ low
6. Hourly logging + trend review

Startup Sequence (Simplified):

- N₂ purge entire system
- Cooling water on
- Deoxo preheat
- Introduce H₂ slowly, monitor reaction
- Bring dryer towers online one by one

Shutdown:

- Stop H₂ feed
- Purge with N₂

- Cool down towers gradually
 - Isolate sections
-

7. Common Problems, Root Causes & Troubleshooting (Real Plant Issues)

1. High Dew Point (Main Complaint):

- Causes: Tower saturated (cycle time short), Valve leakage (wet gas bypass), Cooling failure (water carryover), Desiccant degradation
- Actions: Check tower temps & valve positions, Extend regeneration time, Sample desiccant, Replace if needed

2. High Oxygen:

- Deoxo catalyst poisoned/deactivated, Low temperature, High inlet O₂ load
- Actions: Increase Deoxo temp, Catalyst regeneration or replacement

3. High Pressure Drop:

- Dirty filters, Bed crushing/fouling, Moisture accumulation
- Actions: Backwash or change filters, Inspect bed during shutdown

4. Abnormal Temperatures:

- Heater failure, Thermocouple drift, Valve stuck
- Actions: Calibrate sensors, Check electrical, Manual intervention

5. Frequent Switching / PLC Issues:

- Sensor fault, Logic error
- Actions: Check interlocks, Review alarm history

Desiccant Life: Typically 3–5 years. Poisoning, thermal shock, liquid carryover से कम हो जाता है।

Preventive Maintenance:

- Quarterly: Valve stroke check, Instrument calibration
 - Annual Shutdown: Bed inspection, Desiccant sampling, Cooler cleaning
 - Keep spare desiccant & catalyst
-

8. Optimization & Best Practices

- **Energy Saving:** Use product H₂ for purge & cooling (recycle where possible)
- **Monitoring Trends:** Not just hourly values – look at dew point trend, temperature profiles
- **Safety Interlocks:** High O₂ → trip, High dew point → alarm + switch tower
- **Record Keeping:** Maintain cycle history for each tower (adsorption capacity degradation track)
- **Regeneration Optimization:** Proper heating rate (avoid thermal shock), sufficient purge quantity & time

Key Performance Indicators (KPIs):

- Final Dew Point
 - H₂ consumption (purge loss)
 - Cycle time stability
 - Catalyst/Desiccant life
-

9. Conclusion & Recommendations

यह **Hydrogen Dryer + Deoxo** system safety + quality का backbone है।

- Deoxo → Explosion prevention
- Cooling + Separator → Protection of dryer
- 3-Tower TSA → Continuous ultra-dry H₂

Plant Engineer के रूप में focus:

- Daily log sheet analysis (not just recording)
- Early detection of deviations
- Proper regeneration = longer desiccant life + better performance
- Safety never compromise

Further Reading/Improvement:

- Detailed P&ID study
- PLC logic & sequence diagram (on request)
- Valve sequence table (tower switching)

✓ **Hydrogen Dryer + Deoxo Unit PLC Logic + Valve Sequence + Full Engineer-Level Working Plant Training Document – Operator से Engineer Level तक (Complete, No Topic Skipped – Practical + Technical Detail)**

Prepared for: Plant Engineers, Senior Operators & Maintenance Team **Version:** PLC & Sequence Mastery (Full Detail) **Date:** April 2026

1. COMPLETE PROCESS FLOW (REAL PLANT LOGIC)

Raw Hydrogen (O_2 + Moisture वाला) → **Deoxo Unit** → **Water Cooler** → **Moisture Separator** → **3-Tower Dryer System (T1, T2, T3)** → **Dry Pure Hydrogen** (Dew Point $-60^\circ C$ या नीचे, $O_2 < 10-20$ ppm)

Practical Plant Logic:

- Deoxo में O_2 को permanently हटा दिया जाता है (explosion safety)
- Cooler + Separator moisture का bulk हटाता है
- Dryer towers final polishing करते हैं (deep drying)
- पूरा flow **continuous** रहना चाहिए – कोई भी section fail हुआ तो system trip या bypass logic activate होता है।

Engineer Tip: Flow diagram हमेशा mind में रखो। अगर Deoxo fail → Dryer desiccant तुरंत खराब। अगर Separator fail → Tower life 50% कम।

2. PLC LOGIC – CORE CONCEPT (Brain of the System)

PLC का मुख्य काम (4 चीजें):

- Tower Switching** – Auto cycle चलाना
- Valve Control** – सही समय पर open/close
- Safety Interlocks** – Dangerous condition में system रोकना
- Timing & Sequencing** – Exact timer के हिसाब से step change

3-Tower System का Logic: PLC हमेशा **एक tower Drying** में रखता है ताकि production कभी stop न हो। बाकी दो towers regeneration (heating + cooling) में होते हैं।

Typical Timer Settings (Plant Specific – Example):

- Drying Time = 4 घंटे (main adsorption phase)

- Heating Time = 3 घंटे (regeneration)
- Cooling Time = 1 घंटा (tower को ready करने के लिए) **Total Cycle Time = 8 घंटे** (हर tower 8 घंटे बाद फिर drying में आता है)

PLC Cycle Rotation: PLC timer complete होने पर **Rotate** करता है।

- Step 1: T1 Drying | T2 Heating | T3 Cooling
- Step 2: T2 Drying | T3 Heating | T1 Cooling
- Step 3: T3 Drying | T1 Heating | T2 Cooling फिर वापस Step 1 → 24x7 continuous loop।

PLC Programming Logic (High Level):

- Master Timer ON → Current sequence start
- Drying Timer Complete → “Switch Command” generate
- Safety check (O₂, Dew Point, Pressure) pass → Valves change
- Fail → Sequence “Hold” + Alarm

Engineer Note: PLC में “First Order” और “Last Order” logic होता है – valve close पहले, open बाद में (sequence error रोकने के लिए)।

3. VALVE BY VALVE SEQUENCE (सबसे Important Part)

हर tower में **5 मुख्य Valves** होते हैं (Automatic + Pneumatic/Solenoid operated):

- **IV** = Inlet Valve (Wet H₂ inlet)
- **OV** = Outlet Valve (Dry H₂ outlet)
- **HV** = Heating Valve (Hot purge gas inlet)
- **CV** = Cooling Valve (Cold dry gas inlet)
- **PV** = Purge Valve (Moisture निकलने का outlet)

(A) DRYING MODE – Valve Logic (Production Mode)

Example: Tower-1 Drying

Valve	Status	Reason
T1 Inlet (IV)	OPEN	Wet gas अंदर आए

Valve	Status	Reason
T1 Outlet (OV)	OPEN	Dry gas product line में जाए
T1 Heating (HV)	CLOSED	Heating नहीं चले
T1 Cooling (CV)	CLOSED	Cooling नहीं चले
T1 Purge (PV)	CLOSED	Purge gas बाहर न निकले

बाकी Towers (T2 & T3): Inlet + Outlet **CLOSED** रखे जाते हैं ताकि flow isolate रहे।

(B) HEATING MODE – Regeneration Valve Logic

Example: Tower-2 Heating

Valve	Status	Reason
T2 Inlet (IV)	CLOSED	Wet gas न आए
T2 Outlet (OV)	CLOSED	Dry gas न निकले
T2 Heating (HV)	OPEN	Hot purge gas अंदर आए
T2 Cooling (CV)	CLOSED	Cooling अभी नहीं
T2 Purge (PV)	OPEN	Desorbed moisture purge line से बाहर जाए

क्या होता है? Hot dry gas (product H_2 या N_2 slipstream, 200–300°C) tower से गुजरता है → adsorbed water vaporize होकर purge line से vent/drain में चला जाता है। **Direction:** Counter-current (bottom to top) – best regeneration के लिए।

(C) COOLING MODE – Valve Logic

Example: Tower-3 Cooling

Valve	Status	Reason
T3 Inlet (IV)	CLOSED	Wet gas न आए
T3 Outlet (OV)	CLOSED	Dry gas न निकले
T3 Heating (HV)	CLOSED	Heating बंद
T3 Cooling (CV)	OPEN	Cold dry gas अंदर आए
T3 Purge (PV)	OPEN	Purge gas (heat + residual moisture) बाहर निकले

क्या होता है? Cold dry gas (product H₂ slip) tower को 40–60°C तक ठंडा करता है। Hot tower drying नहीं कर सकता – इसलिए cooling जरूरी।

Important Practical Point: Heating और ON एक ही नहीं हैं।

- ON = Drying mode
 - Heating = Regeneration (tower offline)
-

4. COMPLETE PLC SEQUENCE (STEP-BY-STEP Cycle)

Full Cycle Example (8-Hour Cycle):

● **Step 1 (0–4 hr):**

- T1 → Drying (IV+OV Open)
- T2 → Heating (HV+PV Open)
- T3 → Cooling (CV+PV Open)

● **Step 2 (4–7 hr):** Timer complete → PLC command

- T2 → Drying (IV+OV Open)
- T3 → Heating (HV+PV Open)
- T1 → Cooling (CV+PV Open)

● **Step 3 (7–8 hr):**

- T3 → Drying (IV+OV Open)
- T1 → Heating (HV+PV Open)
- T2 → Cooling (CV+PV Open)

Cycle Repeat → Step 1 वापस। PLC हर switch पर **Valve Sequence** follow करता है:

1. Old drying tower के IV+OV close
2. New drying tower के IV+OV open
3. Regeneration valves accordingly change

Depressurization/Equalization Step: कुछ plants में extra 30–60 sec का step होता है ताकि pressure shock न हो।

5. DEOXO PLC LOGIC (Safety Critical)

Conditions PLC Monitor करता है:

- Deoxo Temperature > 150°C (reaction start)
- O₂ Analyzer < Pre-set limit (normally <20 ppm)
- Heater status + Flow

PLC Actions:

- IF Temperature LOW → Alarm + Heater ON
 - IF O₂ HIGH → High priority Alarm + Possible system Trip / Dryer Bypass
 - IF Heater Fail → Entire system Stop + N₂ Purge
 - Temperature excursion (बहुत high) → Heater OFF + Alarm
-

6. INTERLOCKS (Safety Logic – Non-Negotiable)

Critical Interlocks PLC में Hardwired/Software दोनों:

- High Oxygen → Dryer inlet trip / Alarm + Operator notification
- High Dew Point → Early tower switch + Alarm
- Low H₂ Pressure → System trip (explosion risk)
- High Tower Temperature → Heater OFF
- Valve Fail Feedback (limit switch) → Sequence Hold + Alarm
- Cooling Water Low Flow → Alarm (carryover risk)

Engineer Tip: Interlock bypass केवल maintenance के समय authorized person द्वारा, logbook में entry के साथ।

7. INSTRUMENT WORKING (Log Sheet Values)

- **Pressure Transmitter:** H₂ inlet, N₂ purge, Differential pressure across towers (bed health)
- **Thermocouple:** Deoxo Top/Bottom, Every tower Top/Middle/Bottom (temperature profile)
- **Dew Point Analyzer:** Final outlet – सबसे important quality parameter
- **Oxygen Analyzer:** Post-Deoxo – safety parameter

- **Flow Meters:** H₂ consumption, Purge flow (optimization के लिए)

Log Sheet Reading Skill: Tower temperatures से तुरंत पता चलता है कौन tower किस mode में है।
Dew point rising trend → Regeneration time बढ़ाओ या desiccant check करो।

8. ENGINEERING LOGIC (WHY EVERYTHING HAPPENS)

- **Drying क्यों?** Moisture adsorb → Corrosion रोकना + Product quality
 - **Heating क्यों?** Desiccant को regenerate (reuse) – बिना regeneration tower useless
 - **Cooling क्यों?** Hot desiccant moisture को अच्छे से adsorb नहीं करता
 - **Deoxo क्यों?** O₂ हटाना = Explosion prevention + Dryer protection
 - **3-Tower क्यों?** Continuous production (2-tower में regeneration के दौरान stoppage)
-

9. REAL PLANT TROUBLESHOOTING (Practical)

Problem 1: Dew Point High कारण: Tower saturated / Valve leakage / Water carryover

Solution: Regeneration time check, Valve leak test, Desiccant sample

Problem 2: Oxygen High कारण: Deoxo temp low / Catalyst poisoned Solution: Temp increase / Catalyst regeneration/replacement

Problem 3: Tower Temp Low (Heating Mode) कारण: Heater failure / Thermocouple drift

Solution: Heater check + Calibration

Problem 4: High Pressure Drop कारण: Filter choke / Bed fouling Solution: Filter change + Bed inspection

Problem 5: Valve Not Switching कारण: PLC output fault / Solenoid fail / Air pressure low

Solution: Solenoid test + PLC output check

10. IMPORTANT PRACTICAL POINTS (Plant Engineer Golden Rules)

- Heating और ON **same नहीं** हैं
- Dew Point **सबसे महत्वपूर्ण** parameter है (hourly trend plot करो)
- Tower switching **smooth** होना चाहिए – sudden pressure spike मत होने दो
- Deoxo failure = **Immediate Danger** ⚠
- Cooling mode **skip** नहीं कर सकते

- Regeneration gas consumption 10–20% तक रखो (optimization)
 - Desiccant life 3–5 साल – liquid carryover सबसे बड़ा killer
-

FINAL SIMPLE UNDERSTANDING

पूरा system **एक continuous cycle** है: Remove Oxygen (Deoxo) → Remove Bulk Water (Cooler + Separator) → Deep Dry (Tower Drying) → Regenerate (Heating) → Cool (Ready for next cycle) → Repeat 24×7

PLC इस पूरे dance को perfectly time और safe रखता है।

Hydrogen Dryer + Deoxo Unit Valve Types & Their Real Field Working Full Engineer-Level Training Document (Plant Training Style – Operator से Senior Engineer तक)

Prepared for: Field Operators, Instrumentation Technicians & Plant Engineers **Version:** Valve Mastery (Practical + Technical Detail) **Date:** April 2026

1. Introduction to Valves in Hydrogen Dryer + Deoxo System

Hydrogen Dryer system में valves **बहुत critical** हैं क्योंकि:

- System **continuous** चलता है (24×7)
- High pressure, high temperature, flammable Hydrogen gas
- Automatic tower switching (Drying ↔ Heating ↔ Cooling)
- Safety interlocks depend on correct valve operation

तीन मुख्य valve types जो आपने बताया (Solenoid, Angle, 3-Way) इनका **combined** use होता है। PLC brain है, solenoid signal देता है, और angle/3-way valves actual gas flow control करते हैं।

2. SOLENOID VALVE (Control का Brain – Pilot Valve)

Working Principle (Step-by-Step):

1. PLC output signal (24V DC या 110V AC) solenoid coil को energize करता है।
2. Coil magnetic field create करता है।
3. Plunger (armature) move होता है।
4. Plunger air port open/close करता है।

5. Instrument air solenoid valve के जरिए main pneumatic actuator को supply देता है या cut करता है।

Types in Plant:

- 3/2 Way Solenoid (most common)
- 5/2 Way Solenoid (for double acting actuator)
- Direct acting या Pilot operated

Real Use in Dryer System:

- हर pneumatic valve (angle या 3-way) के actuator को control करने के लिए
- Tower switching sequence में सभी valves को synchronized तरीके से open/close करने के लिए
- Safety interlock valves (emergency shut-off)

PLC Connection: PLC Digital Output → Solenoid Coil → Feedback (limit switch) PLC input में जाता है।

Why Very Important?

- Fast response (milliseconds)
- Remote automatic control
- PLC logic के बिना manual operation मुश्किल

Common Field Problems & Troubleshooting:

- Coil burn (overvoltage या continuous energize) → Resistance check करो (normal 20–100 ohm)
- No air supply → Instrument air pressure low (min 4–6 bar)
- Valve stuck (dirt, corrosion) → Clean या replace
- No feedback signal → Limit switch faulty

Engineer Tip: Solenoid failure = Tower switching fail → Dew point high → Immediate alarm। हमेशा spare solenoid रखो।

3. ANGLE VALVE (Heavy Duty Flow Control Valve)

Working Principle:

- Body design 90° angle में होता है (inlet bottom, outlet side)

- Flow path smooth, pressure drop कम
- Actuator: Pneumatic (diaphragm या piston type)
- Solenoid valve इस actuator को air supply देता है → stem move → plug-seat open/close

Key Features:

- High pressure & temperature rating (up to 300°C+)
- Tight shut-off (Class VI leakage class often)
- Good for gas service (Hydrogen)

Real Use in Dryer System:

- Main H₂ inlet line
- Heating gas line (hot purge gas)
- Purge outlet line
- Outlet dry H₂ line
- Isolation valves जहाँ high flow या high temp हो

Why जरूरी?

- Hydrogen जैसा light gas + high temp में normal globe valve से बेहतर performance
- Angle design में dirt accumulation कम होता है
- Fast opening/closing with pneumatic actuator

Common Field Problems:

- Seat damage (high velocity gas erosion या moisture) → Leakage across valve
- Actuator diaphragm tear → Valve not fully open/close
- Stem packing leakage (Hydrogen escapes → safety hazard)
- Positioner failure (if smart valve) → Wrong position

Maintenance Tip: Angle valves को yearly shutdown में overhaul करो। Seat & plug inspect करो। Hydrogen service में special packing material (Graphite या PTFE) use होता है।

4. 3-WAY VALVE (Direction Changer Master)

Working Principle: 3 ports वाले valve:

- Common Port (Inlet)
- Port A
- Port B

Two Main Types in Dryer:

A. Diverting Type (most common in dryer):

- 1 Inlet → 2 Outlets
- Gas को Tower-1 या Tower-2 की तरफ divert करता है

B. Mixing Type (less common):

- 2 Inlets → 1 Outlet

Actuation: Pneumatic actuator + Solenoid control Position feedback (limit switches या position transmitter) PLC को जाता है।

Real Use in Dryer System (सबसे Critical):

- Inlet gas switching: Wet H₂ को कौन से tower में भेजना है (T1 → T2 → T3)
- Purge gas routing
- Cooling gas diversion
- Regeneration gas selection

Example in Operation:

- Drying mode: 3-way valve gas को Tower-1 की तरफ divert करता है
- Tower switch के समय: 3-way valve position change → Gas अब Tower-2 में जाता है
- Heating mode: 3-way valve hot gas को regenerating tower की तरफ भेजता है

Why जरूरी?

- बिना 3-way valve के tower switching के लिए 4-6 अलग valves लगाने पड़ते (complex piping)
- Smooth & fast direction change
- Continuous flow maintain करता है (no production stop)

Common Field Problems:

- Internal leakage (between ports) → Wet gas dry tower में leak → Dew point suddenly high

- Wrong switching (actuator fail) → Wrong tower active
- Position feedback error → PLC sequence hold या alarm
- Ball/disc sticking (moisture या temperature cycle से)

Engineer Tip: 3-way valves को leakage test (bubble test या helium leak test) regularly करो। Hydrogen में leakage बहुत dangerous।

5. COMBINED WORKING – Real Process Logic (How They Work Together)

PLC → Solenoid → Pneumatic Actuator → Main Valve (Angle or 3-Way)

Example 1: Tower-1 Drying Mode

1. PLC command → Solenoid valves energize
2. Solenoid → Air to 3-way Inlet Valve → Gas Tower-1 में divert
3. Solenoid → Air to Angle Inlet Valve → Open
4. Solenoid → Air to Angle Outlet Valve → Open
5. Heating & Cooling valves के solenoids de-energize → Closed

Example 2: Tower-2 Heating (Regeneration) Mode

1. PLC → Solenoid for Heating Valve (Angle or 3-way) → OPEN
2. Hot purge gas line open
3. Purge Valve (Angle) open → Moisture बाहर निकले
4. Inlet & Outlet valves closed (3-way + Angle combination)

Example 3: Tower-3 Cooling Mode

1. PLC → Cooling 3-way valve position change
2. Cold dry gas Tower-3 में divert
3. Purge valve open → Hot gas निकले

Sequence Safety: PLC हमेशा **Fail-safe sequence** follow करता है:

- पहले old valves close
- Pressure equalize (few seconds)
- New valves open

- Feedback confirm → Next step

6. REAL FIELD FAULTS & Troubleshooting (Operator Level)

Problem	Most Likely Reason	Immediate Action	Preventive Fix
Tower not switching	3-way valve stuck or actuator fail	Manual override (if safe) + Alarm	Clean / Overhaul v
Valve not opening	Solenoid coil fail or no air supply	Check solenoid LED + Air pressure	Replace solenoid, dryer
Leakage across closed valve	Angle valve seat damage	Isolate & monitor dew point	Replace seat & pl
Wrong flow direction	3-way valve internal leakage or miswired	Check position feedback	Leak test + recalib
Slow valve operation	Low instrument air pressure / clogged line	Increase air pressure	Clean air lines + fi
Frequent solenoid failure	Moisture in air or overvoltage	Check air quality	Install better air d

Golden Rule: Dew point suddenly bad → 90% chance valve related problem (leakage या wrong switching)

7. SIMPLE ONE-LINE UNDERSTANDING

- **Solenoid Valve** = Brain (PLC का electrical signal giver)
- **Angle Valve** = Muscle (Heavy duty flow on/off & control)
- **3-Way Valve** = Direction Master (Tower switching का hero)

Complete Chain: PLC Logic → Digital Output → Solenoid Valve → Instrument Air → Pneumatic Actuator → Angle/3-Way Valve → Hydrogen Gas Flow Control

Document Complete

अब बताओ Devanand, आपको next क्या चाहिए?

1. **P&ID Diagram with Valve Tagging** (XV-101, SV-201, 3WV-301 etc. के साथ text-based या descriptive)
2. **Valve Troubleshooting Checklist** (daily/weekly/monthly operator level)
3. **Real Plant Case Studies** (dew point high due to 3-way valve leakage जैसी stories)
4. **Ladder Logic Example** for one tower switching sequence

या कोई specific valve का और deep detail चाहिए (actuator types, positioners, leakage class आदि)।

बोलो – मैं exact वही बना दूँगा जो plant floor पर training में इस्तेमाल होता है! 🔥

1. Introduction – Why Valves are Critical in Hydrogen Dryer + Deoxo System

Hydrogen Dryer + Deoxo system में valves **system की रीढ़** हैं क्योंकि:

- Hydrogen flammable और explosive gas है (LEL 4%) → कोई leakage या wrong switching = safety hazard
- System **continuous 24x7** चलता है – production stop नहीं हो सकता
- Temperature Swing Adsorption (TSA) में towers हर 4-8 घंटे switch होते हैं (Drying → Heating → Cooling)
- High temperature (Deoxo 150-300°C, Regeneration 200-315°C), high pressure, और moisture present
- PLC fully automatic control करता है – manual intervention minimum

तीन मुख्य valve types (जो आपने बताया): **Solenoid, Angle, और 3-Way**। ये तीनों **combined** काम करते हैं: PLC brain → Solenoid signal → Instrument air → Pneumatic actuator → Angle/3-Way valve → Hydrogen gas flow control

P&ID में Symbols (Standard ISA):

- Solenoid Valve: Rectangle with coil symbol + arrow for flow
- Angle Valve: Globe valve जैसा लेकिन 90° angle flow path
- 3-Way Valve: Three triangles या T-shape with arrows showing diverting/mixing

2. SOLENOID VALVE (Control का Boss – PLC Operated Pilot Valve)

Detailed Working Principle (Step-by-Step):

1. PLC Digital Output (24V DC सामान्यतः) solenoid coil को energize करता है।
2. Coil में current flow → Magnetic field generate होता है।
3. Magnetic field plunger (armature) को attract/repel करता है।
4. Plunger move → Internal pilot port open/close होता है।
5. Instrument air (4-7 bar, dry & clean) main pneumatic actuator को supply मिलता है या exhaust हो जाता है।
6. Actuator stem move → Main valve (Angle या 3-Way) open/close या position change।

Types Commonly Used:

- 3/2 Way (3 ports, 2 positions) – most common for single acting actuator
- 5/2 Way – double acting actuator के लिए
- Direct Acting (small valves) या Pilot Operated (large flow)
- Normally Closed (NC) या Normally Open (NO) – fail-safe design के अनुसार

Real Use in Dryer System:

- हर pneumatic main valve के pilot के रूप में
- Tower switching sequence (Drying/Heating/Cooling valves)
- Emergency isolation valves
- Purge, heating & cooling line control

PLC Connection:

- PLC Output → Solenoid Coil
- Feedback: Limit switch या position transmitter → PLC Input (valve fully open/closed confirm)
- Interlock: Valve feedback mismatch → Sequence hold + Alarm

Why जरूरी?

- Fast response (milliseconds)
- Remote & automatic operation
- Precise timing for tower cycle (PLC timer के साथ sync)
- Safety: Fail-safe position (de-energized पर safe state)

Common Field Problems & Root Causes + Fixes:

- **Coil Burn:** Overvoltage, continuous energize, high ambient temp → Coil resistance check (20-100 ohm normal), replace coil
- **No Air Supply:** Instrument air dryer fail, low pressure (<4 bar), clogged filter → Air quality check (ISO 8573), drain moisture
- **Valve Stuck:** Dirt, corrosion, moisture in air → Clean plunger/orifice, replace if damaged
- **Not Opening/Closing:** Wrong voltage, wiring fault, debris → Multimeter से continuity & voltage check
- **Slow Operation:** Low air pressure or restricted exhaust → Air line cleaning

Engineer Tip: Hydrogen service में solenoid body stainless steel (SS316) और seals FKM/PTFE use होते हैं। Spare solenoid हर valve के साथ रखो।

3. ANGLE VALVE (Heavy Duty Flow Control Valve)

Detailed Working Principle:

- Body design: Inlet bottom से, outlet 90° side में – flow path angle में change होता है
- Lower pressure drop compared to globe valve
- Actuator: Pneumatic diaphragm या piston type (solenoid pilot से control)
- Stem move → Plug-seat contact → Flow open/close या throttle
- Tight shut-off (Class IV to VI leakage class)

Key Features for Hydrogen Service:

- High temperature rating (up to 300°C+)
- High pressure handling
- Smooth flow – erosion कम
- Packing: Graphite or special PTFE for low hydrogen leakage

Real Use in Dryer System:

- Main H₂ Inlet line (isolation)
- Heating line (hot purge gas)
- Purge outlet line (moisture vent)

- Dry H₂ Outlet line
- Isolation valves जहाँ high flow/temp हो

Why जरूरी?

- Angle design में fewer pipe fittings लगते हैं
- Hydrogen + high temp में reliable tight shut-off
- Moisture या particles handle करने में बेहतर

Common Field Problems & Troubleshooting:

- **Seat Damage/Leakage:** High velocity erosion, thermal cycling, moisture corrosion → Bubble test या helium leak test, seat & plug replace
- **Actuator Fail:** Diaphragm tear, spring broken → Actuator overhaul
- **Stem Packing Leakage:** Hydrogen escapes (dangerous) → Repack with correct material
- **Slow or Jerky Movement:** Air supply issue या positioner fault → Calibrate positioner (if smart valve)
- **High Pressure Drop:** Partial blockage → Clean strainer upstream

Maintenance Tip: Yearly shutdown में angle valves को full overhaul करो। Hydrogen embrittlement check करो (material brittleness)।

4. 3-WAY VALVE (Direction Change Master – Tower Switching Hero)

Detailed Working Principle:

- 3 Ports: Common (Inlet), Port A, Port B
- Actuator (pneumatic) position change करता है → Flow direction divert या mix
- Types:
 - **Diverting Type** (1 Inlet → 2 Outlets) – Dryer में सबसे common
 - **Mixing Type** (2 Inlets → 1 Outlet) – कम use

Operation:

- De-energized position: Flow Common → Port A
- Energized (via solenoid): Flow Common → Port B

- Position feedback (2 limit switches या 4-20mA transmitter) PLC को confirm भेजता है

Real Use in Dryer System (सबसे Important):

- Inlet gas switching: Wet H₂ को T1, T2 या T3 tower में divert करना
- Regeneration gas routing (heating gas कौन से tower में)
- Cooling gas diversion
- Purge line selection

Practical Example:

- Drying mode: 3-way valve gas को Tower-1 की तरफ divert
- Tower switch command: 3-way valve position change → Gas अब Tower-2 में जाता है (continuous flow)

Why जरूरी?

- Multiple 2-way valves की जगह एक 3-way valve से piping simple
- Fast & smooth tower selection
- Continuous production maintain (no flow interruption)

Common Field Problems & Troubleshooting:

- **Internal Leakage (Port to Port):** Wet gas dry tower में leak → Dew point suddenly high → Pressure test या leak detection
- **Wrong Switching:** Actuator fail, solenoid issue, miswiring → Position feedback check
- **Stuck Position:** Thermal cycling, debris, moisture → Manual exercise (safe condition में), clean or overhaul
- **Position Feedback Error:** Limit switch drift → Recalibrate or replace
- **High Torque Required:** Actuator undersized → Check actuator sizing

Engineer Tip: Molecular sieve switching valves के लिए tight bidirectional shut-off जरूरी। Ball-type या special trunnion mounted 3-way valves अच्छे perform करते हैं।

5. COMBINED WORKING – Real Process Examples (How All Three Work Together)

PLC → Solenoid → Air → Actuator → Main Valve (Angle or 3-Way)

Example 1: Tower-1 Drying Mode

- PLC command → Inlet 3-way valve solenoid ON → Gas Tower-1 में divert
- Angle Inlet Valve solenoid → Open (flow allow)
- Angle Outlet Valve → Open
- Heating & Cooling valves closed (their solenoids de-energized)

Example 2: Tower-2 Heating (Regeneration) Mode

- PLC → Heating line 3-way/Angle valve solenoid ON
- Hot purge gas (200-300°C) Tower-2 में pass
- Purge Angle Valve open → Desorbed moisture बाहर निकले

Example 3: Tower-3 Cooling Mode

- PLC → Cooling 3-way valve position change
- Cold dry H₂ slip Tower-3 में divert
- Purge valve open → Residual heat निकले

Valve Switching Sequence Safety (PLC Logic):

1. Old drying tower valves close first
2. Short depressurization/equalization time (few seconds)
3. New valves open
4. Feedback confirm → Proceed
5. Mismatch → Alarm + Hold

6. REAL FIELD FAULTS & Troubleshooting Table (Operator Level)

Problem	Likely Reason	Immediate Action	Long-term Fix
Tower change नहीं हो रहा	3-way valve stuck / actuator fail	Check air pressure, manual override (safe)	Overhaul valve, clean actuator
Any valve open नहीं हो रहा	Solenoid fail / no air / coil burn	Check solenoid LED/voltage, air supply	Replace solenoid + air dryer check
Leakage (closed valve)	Angle valve seat damage	Monitor dew point, isolate if possible	Replace seat/plug, leak test

Problem	Likely Reason	Immediate Action	Long-term Fix
Wrong flow direction	3-way internal leak / misconnection	Check position feedback	Leak test + recalibrate
Slow valve movement	Low air pressure / clogged lines	Boost air pressure	Clean instrument air system
Frequent solenoid failure	Moisture in air / overvoltage	Check air quality	Install better coalescing filter
High dew point after switch	Valve leakage (3-way or angle)	Valve leak test	Repair/replace switching valves

Golden Rule for Field: Dew point खराब → 80-90% chance valve related (leakage या wrong position)। तुरंत tower temps + valve feedback check करो।

7. SIMPLE ONE-LINE UNDERSTANDING

- **Solenoid Valve** = Brain (PLC electrical signal → Air pilot)
- **Angle Valve** = Muscle (Heavy duty on/off & flow control at 90°)
- **3-Way Valve** = Direction Changer (Tower selection & flow routing master)

Complete Control Chain: PLC Logic → Solenoid Valve → Instrument Air → Pneumatic Actuator → Angle Valve or 3-Way Valve → Precise Hydrogen Gas Flow & Tower Switching

Document Complete (Plant Floor Training Ready)

अब बताओ Devanand, next क्या चाहिए?

1. **P&ID Diagram with Valve Tagging** (XV-101 for angle, SV-201 for solenoid, 3WV-301 for 3-way आदि के साथ descriptive)
2. **Daily/Weekly Troubleshooting Checklist** (operator level)
3. **Real Plant Case Studies** (e.g., 3-way valve leakage से dew point high)
4. **Ladder Logic Snippet** for valve switching sequence

या किसी valve का और deep technical detail (materials, leakage class, sizing आदि) चाहिए।

बोलो – मैं exact plant training material की तरह next part बना दूँगा! 🔥